$E(X^r) = M_x^r(t)$ at t=0 $M_x^r(t)$

r^{th} differential of $M_x(t)$ is

$\mu_2 = \mu_2' - (\mu_1')^2$

To find the rth moment of X about the origin.

We know that

$$M_x(t) = E(e^{tX})$$

$$= E\left[1 + \frac{tX}{1!} + \frac{(tX)^2}{2!} + \frac{(tX)^3}{3!} + \ldots + \frac{(tX)^r}{r!} + \ldots\right]$$

$$= 1 + \frac{E(tX)}{1!} + \frac{E(tX)^2}{2!} + \frac{E(tX)^3}{3!} + \ldots + \frac{E(tX)^r}{r!} + \ldots$$

$$= 1 + t\frac{E(X)}{1!} + \frac{t^2E(X^2)}{2!} + \ldots + \frac{t^rE(X^r)}{r!} + \ldots$$

$$= 1 + t\mu'_1 + t^2\frac{\mu'_2}{2!} + \ldots + t^r\frac{\mu'_r}{r!} + \ldots$$

$$M_x(t) = \sum_{r=0}^{\infty} t^r \frac{\mu'_r}{r!}$$

From the series on the right hand side, μ_r' is the coefficient of $\frac{t^r}{r!}$ in $M_x(t)$.

Since $M_x(t)$ generates moments of the distribution and hence it is known as moment generating function.

Using the function, we can find mean and variance by using the first two raw moments.

$\mu_2 = \mu'_2 - (\mu'_1)^2$

9.8.4. Characteristic function

For some distribution, the M.G.F does not exist. In such cases we can use the characteristic function and it is more servicable function than the M.G.F.

Definition:

The characteristic function of a random variable X, denoted by $\phi_x(t)$, where $\phi_x(t) = E(e^{i\,t\,X})$ then

$$\phi_x(t) = E(e^{i\,tX}) = \int_{-\infty}^{\infty} e^{itx} f(x)\,dx \text{, for continuous random variable}$$

$$\phi_x(t) = \sum_e^{itx} p(x) \text{ , for discrete random variable}$$

where $i = \sqrt{-1}$